SAFETY CCTV CLIENT MANUAL

AI-Powered Workplace Safety, Danger Zone Intrusion & PPE Compliance Suite

Target Environment: Windows 10 / 11 (64-bit)	**Runtime:** Standalone CPU-Optimized (No Python needed)
Interface: Local Web Dashboard (Port 8000)	**Deployment Status:** Production Standalone Release

1. System Overview & Core Capabilities

Safety CCTV is an offline computer vision monitoring system engineered to run on standard client laptops and workstations. Using optimized neural network models running directly on the host CPU, the application continuously monitors video streams to prevent workplace accidents and ensure regulatory compliance.

The software provides three distinct operational stations:

1. Zone Alert	Monitors designated restricted danger zones. Fires an audio siren upon intrusion, tags the intruder ID, and preserves timestamped evidence snapshots in evidence/.
2. PPE Mask + Helmet	Entry-gate safety checkpoint checking for mandatory hardhats and facial masks in real time.
3. PPE Helmet + Vest + Gloves	Full 3-point compliance checkpoint combining body tracking, fine-tuned glove detection, and MediaPipe landmark heuristics for active work areas.

2. Quick-Start Guide (Ready in 3 Minutes)

This software requires **zero software installations** on the target laptop. Follow these three simple steps to start monitoring:

Step 1	**Extract & Place Application:** Unzip SafetyCCTV_win64.zip into any folder on your laptop (e.g. C:\SafetyCCTV or Desktop). All dependencies and AI models are contained inside.
Step 2	**Define Danger Zones:** Run tools/DefineZones.exe. Your camera view will open. Click points to outline your danger zone polygon. Press **'S'** to save the zone, then **'Q'** to exit.
Step 3	**Double-Click SafetyCCTV.exe:** The web console automatically launches at http://localhost:8000 in your default web browser. Click **'Start'** on any station.

3. Directory Layout & Components

The software package is organized into clearly defined directories:

Directory / File	Function & Description
`SafetyCCTV.exe`	Main double-click application launcher. Boots backend & opens browser.
`_internal/`	Shared AI runtime and libraries (PyTorch CPU, OpenCV, Ultralytics). Do not edit.
`models/`	Deep learning model weights (ppe_v8n.pt, ppe_ft.pt, yolo26n.onnx).
`config/`	System settings (settings.json), danger zones (zones.json), and alert settings.
`assets/`	Dashboard UI (console.html) and audio siren (alarm.wav).
`tools/`	Standalone utilities: DefineZones.exe, CalibrateGlove.exe, CollectData.exe.
`logs/`	Daily CSV compliance logs recording hourly statistics and incident timestamps.
`reports/`	Formatted Microsoft Excel audit reports generated on-demand from the console.
`evidence/`	Timestamped JPEG violation photos organized automatically by date (YYYY-MM-DD).

4. Hardware & Configuration Guide

Configuring Camera Source (config/settings.json)

By default, the system uses external USB camera index "1". If you are using the laptop's built-in webcam or an RTSP network CCTV stream, open config/settings.json in Notepad and set:

Built-in Laptop Webcam:	`{"camera": "0", "port": 8000}`
External USB Camera:	`{"camera": "1", "port": 8000}`
IP CCTV Network Stream:	`{"camera": "rtsp://admin:pass@192.168.1.100:554/ch1", "port": 8000}`

Note: Any changes saved to settings.json take effect immediately on the next station start without restarting the launcher.

Automated Email & WhatsApp Alerts (Optional)

To receive instant alerts when danger intrusions or safety non-compliance occurs:
1. Copy config/alerts.example.json and rename it to config/alerts.json.
2. Open config/alerts.json in Notepad and enter your credentials:
   • **Email:** Enter your SMTP host, port, username, and password (for Gmail, generate a 16-character App Password).
   • **WhatsApp:** Enter your phone number and free CallMeBot API key.
3. Set "enabled": true. Alerts will automatically trigger with cool-downs to prevent inbox spam.

5. Operating the Web Dashboard & Generating Reports

When SafetyCCTV.exe runs, it serves a real-time reactive console at http://localhost:8000. The interface is optimized for both desktop and wall-mounted monitoring screens, featuring Light/Dark modes, live status indicators, hourly metrics, and an evidence photo wall.

Feature	Operational Details
Station Start / Stop	Only one station uses the camera at any moment. Clicking 'Start' on Station B cleanly terminates Station A, releases the camera lock, and initializes the new neural network within seconds.
Real-time Metrics	Six live counters tally daily statistics: Danger Entries, Total Visitors Seen, No-Helmet Incidents, No-Vest Incidents, No-Mask Incidents, and No-Gloves Incidents.
Live Evidence Wall	Violation photos are captured automatically and displayed as visual thumbnails. Clicking any thumbnail opens the full-resolution high-definition captured frame.
Excel Audit Reports	Click **'Zone Excel'** or **'PPE Excel'** to instantly download complete Microsoft Excel reports compiled from today's CSV logs, broken down by hour and incident type.

6. Troubleshooting & Operational FAQs

Issue Observed	Resolution Steps
Camera open failed / black screen	1. Check if another program (Zoom, Teams, Skype, Windows Camera) is holding the webcam. 2. Verify whether your camera is index "0" or "1" in config/settings.json. 3. Replug the USB cable and click 'Start' again.
Web page says 'Unable to connect'	If port 8000 is occupied by another application, open config/settings.json, change "port": 8000 to "port": 8080, and restart SafetyCCTV.exe.
App opens browser but does not launch twice	This is expected. Safety CCTV has single-instance protection. Double-clicking the icon while already active simply brings up the existing dashboard without creating duplicate processes.
Zone alarms fire unexpectedly	Run tools/DefineZones.exe to verify the polygon boundary. Ensure it only covers restricted floor regions and does not overlap safe pedestrian walkways.
Custom glove colors not recognized	Run tools/CalibrateGlove.exe under standard facility lighting. Hold the glove in the green square and press 'S' to calibrate the color filter into config/glove_hsv.json.

Customer Support & Updates: Safety CCTV is designed for self-contained, air-gapped operation. All data, models, and incident evidence remain strictly confidential on the local computer.